Tekrar

s laplace

$s = j\omega$

$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

z-Tranform

$z = e^{j\omega}$

$X(e^{j\omega})$

Ayrık zamanlı fourier transformu (Dönüşümü)

(Discrete-time Fourier Transform)

FT, periyodik olmayan işaretler için tanımlıdır.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

(n: tamsayı)

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$x[n] \xrightarrow{FF} X(e^{j\omega}), \qquad FF^{-1}$$

⇒ Ayrık zamanlı FT, $X(e^{j\omega})$ 2π ile periyodiktir

$$X(e^{j\omega}) = X(e^{j(\omega + 2\pi)})$$

Ex: $x[n] = a^n u[n]$, $|a| < 1$ ise $X(e^{j\omega}) = ?$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0} (a e^{-j\omega})^n = \frac{1}{1 - a e^{-j\omega}}$$

Ex: $x[n] = a^{|n|}$, $|a| < 1$ ise $X(e^{j\omega}) = ?$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} a^{|n|} e^{-j\omega n} = \sum_{n=-\infty}^{-1} a^{-n} e^{j\omega n} + \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

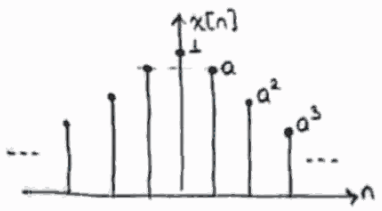
$$\sum_{n=-\infty}^{-1} (a e^{j\omega})^{-n} = \sum_{k=1}^{\infty} (a e^{j\omega})^k = \sum_{k=0}^{\infty} (a e^{j\omega})^k - 1 = \frac{1}{1 - a e^{-j\omega}} - 1 = \frac{a e^{j\omega}}{1 - a e^{j\omega}}$$

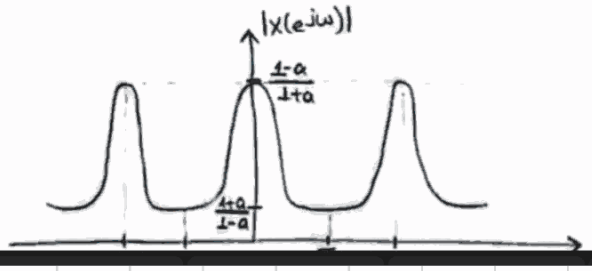
($k = -n$; $(a e^{j\omega})^0 = 1$)

$$X(e^{j\omega}) = \frac{a e^{j\omega}}{1 - a e^{j\omega}} + \frac{1}{1 - a e^{-j\omega}} = \frac{1 - a^2}{1 + a^2 - 2a\cos\omega} \quad : \text{Reel}$$

$X(e^{j\omega}) = |X(e^{j\omega})|$

$\measuredangle X(e^{j\omega}) = 0$





Ex: $x[n]=\begin{cases}1, & |n|\le N_1\\ 0, & |n|>N_1\end{cases}$ şeklinde verilen ayrık zamanlı dikdörtgen darbe işaretinin FT bulunuz.

$$X(e^{jw})=\sum_{n=-\infty}^{\infty}x[n]e^{-jwn}=\sum_{n=-N_1}^{N_1}e^{-jwn}=\sum_{n=0}^{2N_1}e^{-jw(n-N_1)}=e^{jwN_1}\sum_{n=0}^{2N_1}e^{-jwn}$$

$$=e^{jwN_1}\left[\frac{1-e^{-jw(2N_1+1)}}{1-e^{-jw}}\right]$$

Reminder: $\sum_{k=0}^{M}\alpha^k=\frac{1-\alpha^{M+1}}{1-\alpha}$

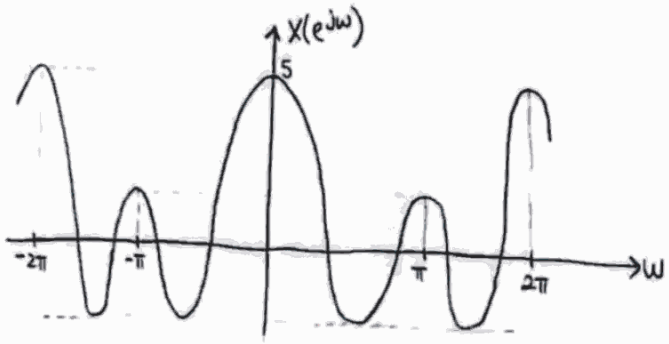
$$1-e^{j\theta}=e^{-j\theta/2}\left[e^{j\theta/2}-e^{-j\theta/2}\right]$$

$$=e^{jwN_1}\frac{e^{-jw(N_1+1/2)}\left[e^{jw(N_1+1/2)}-e^{-jw(N_1+1/2)}\right]}{e^{-jw/2}\left[e^{jw/2}-e^{-jw/2}\right]}$$

$$=\frac{\sin(w(N_1+1/2))}{\sin(w/2)}$$

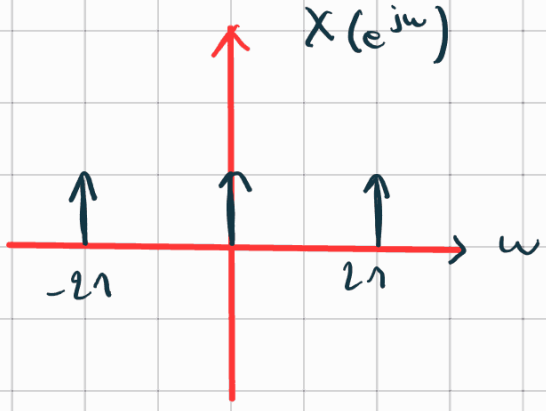
$N_1=2$ için





★ $\delta[n] \overset{F}{\longleftrightarrow} 1$

$1 \longleftrightarrow 2\pi\,\delta(e^{jw})$



$\delta[n-n_0] \longleftrightarrow e^{-jwn_0}$

- $e^{jw_0n} \overset{FF}{\longleftrightarrow} 2\pi\,\delta(e^{j(w-w_0)})$

$e^{-jw_0n} \overset{FF}{\longleftrightarrow} 2\pi\,\delta(e^{j(w+w_0)})$

Periyodik işaretlerin Fourier Transformu

$$x[n] = \sum_{k=\langle N \rangle} x_k e^{jk\omega_0 n} = x_0 + x_1 e^{j\omega_0 n} + x_2 e^{j2\omega_0 n} + \cdots +$$

$$x_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n}, \quad x_k = x_{k+n}$$

$$x[n] = \sum_{k=\langle N \rangle} x_k e^{jk\omega_0 n} \xleftrightarrow{FF} X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi x_k \delta(\omega - k\omega_0)$$

Ayrık Zamanlı Fourier Transformunun Özellikleri

$x[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})$

$X(e^{j\omega}) = \mathcal{F}\{x[n]\}$

$x[n] = \mathcal{F}^{-1}\{X(e^{j\omega})\}$

1) Periyodiklik: $X(e^{j\omega}) = X(e^{j(\omega+2\pi)})$, 2π ile periyodiktir.

2) Lineerlik: $\alpha x_1[n] + \beta x_2[n] \xleftrightarrow{\mathcal{F}} \alpha X_1(e^{j\omega}) + \beta X_2(e^{j\omega})$

3) Zamanda öteleme: $x[n-n_0] \xleftrightarrow{\mathcal{F}} e^{-j\omega n_0} X(e^{j\omega})$

Genlik spektrumu değişmez, faz spektrumu $-\omega n_0$ kadar ötelenir.

4) Frekansta öteleme: $e^{j\omega_0 n} x[n] \xleftrightarrow{\mathcal{F}} X(e^{j(\omega-\omega_0)})$

5) Zamanda katlama: $x[-n] \xleftrightarrow{\mathcal{F}} X(e^{-j\omega})$

6) Eşlenik alma: $x^*[n] \xleftrightarrow{\mathcal{F}} X^*(e^{-j\omega})$

7) Fark alma (differencing): $x[n] - x[n-1] \xleftrightarrow{\mathcal{F}} (1-e^{-j\omega}) X(e^{j\omega})$

7) Fark alma (differencing): $x[n] - x[n-1] \xleftrightarrow{\mathcal{F}} (1-e^{-j\omega}) X(e^{j\omega})$

8) Birikim (Accumulation): $\sum_{m=-\infty}^{n} x[m] \xleftrightarrow{\mathcal{F}} \frac{1}{1-e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$

9) Frekansta türev: $n\,x[n] \xleftrightarrow{\mathcal{F}} j\frac{dX(e^{j\omega})}{d\omega}$

İspat: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$d\omega$

İspat: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

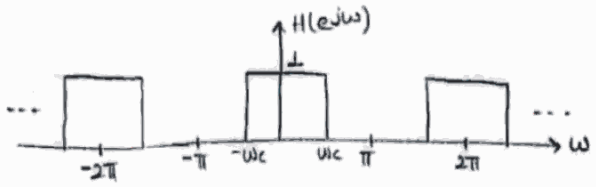
$\frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \frac{d}{d\omega}(e^{-j\omega n}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \cdot (-jn)$

10) Parseval teoremi: $E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$

11) Çarpma: $x_1[n] \cdot x_2[n] \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\alpha}) X_2(e^{j(\omega-\alpha)})\, d\alpha$

konvülasyon

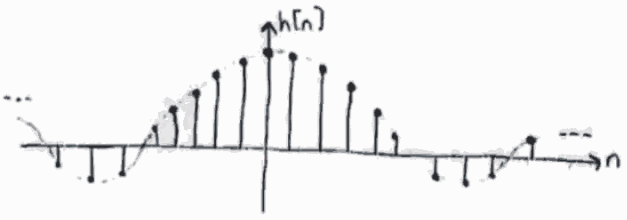
Ex: $H(e^{jw}) = \begin{cases} 1, & 0 \le |w| \le w_c \\ 0, & w_c < |w| < \pi \end{cases}$ ayrık zamanlı alçak geçiren filtrenin dürtü cevabını bulunuz.



$$h[n] = \frac{1}{2\pi}\int_{2\pi} H(e^{jw})\, e^{+jwn}\, dw = \frac{1}{2\pi}\int_{-w_c}^{w_c} e^{jwn}\, dw = \frac{1}{2\pi}\,\frac{e^{jwn}}{jn}\Big|_{-w_c}^{w_c} = \frac{1}{2j\pi n}\left[e^{jw_c n} - e^{-jw_c n}\right]$$

$$h[n] = \frac{1}{\pi n}\sin(w_c n) = \frac{\sin\left(\pi \frac{w_c}{\pi} n\right)}{\pi \frac{w_c}{\pi} n}\,\frac{w_c}{\pi} = \frac{w_c}{\pi}\,\mathrm{sinc}\left(\frac{w_c}{\pi} n\right)$$

↳ tam sayı olunca 0'ı yani n eksenini keser.



Ex:

$$x[n] = \alpha^n u[n],\ |\alpha| < 1,\ g[n] = ?$$
$$h[n] = \beta^n u[n],\ |\beta| < 1$$

Reminder: $\alpha^n u[n] \xleftrightarrow{FF} \frac{1}{1-\alpha e^{-jw}}$

$$Y(e^{jw}) = \frac{1}{(1-\alpha e^{-jw})(1-\beta e^{-jw})} = \frac{A}{1-\alpha e^{-jw}} + \frac{B}{1-\beta e^{-jw}}$$

$\updownarrow FF^{-1}$

$$y[n] = A\alpha^n u[n] + B\beta^n u[n]$$

$$A = \frac{\alpha}{\alpha-\beta} \qquad B = \frac{-\beta}{\alpha-\beta}$$

$$y[n] = \frac{1}{\alpha-\beta}\left[\alpha\,\alpha^n u[n] - \beta\,\beta^n u[n]\right]$$

$$= \frac{1}{\alpha-\beta}\left(\alpha^{n+1} - \beta^{n+1}\right)u[n]$$

Ex: $\alpha = \beta$ ise $y[n] = ??$

$$Y(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})^2}$$

Reminder:

$$x[n] = \alpha^n u[n] \xleftrightarrow{FF} X(e^{j\omega}) = \frac{1}{1-\alpha e^{-j\omega}}$$

$$n\,x[n] \xleftrightarrow{FF} j\frac{dX(e^{j\omega})}{d\omega}$$

$$x[n-n_0] \xleftrightarrow{FF} e^{-j\omega n_0} X(e^{j\omega})$$

$$\frac{dX(e^{j\omega})}{d\omega} = -j\alpha e^{-j\omega}\frac{1}{(1-\alpha e^{-j\omega})^2} = -j\alpha e^{-j\omega} Y(e^{j\omega})$$

$$Y(e^{j\omega}) = \frac{1}{\alpha} e^{j\omega} j\frac{dX(e^{j\omega})}{d\omega} \xleftrightarrow{FF^{-1}} y[n] = \frac{1}{\alpha}(n+1)\alpha^{n+1}u[n+1]$$

$$= (n+1)\alpha^n u[n]$$

$$y[-1] = 0$$

$$(n+1)\alpha^n u[n] \xleftrightarrow{FF} \frac{1}{(1-\alpha e^{-j\omega})^2}$$

Ex: $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$ ise $h[n] = ?$

$$Y(e^{j\omega}) - \frac{3}{4}e^{-j\omega}Y(e^{j\omega}) + \frac{1}{8}e^{-j2\omega}Y(e^{j\omega}) = 2X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{2}{1-\frac{3}{4}e^{-j\omega}+\frac{1}{8}e^{-j2\omega}} = \frac{A}{1-\frac{1}{2}e^{-j\omega}} + \frac{B}{1-\frac{1}{4}e^{-j\omega}}$$

FF^{-1}

$A = 4 \qquad B = -2$

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n]$$

Laplace Transformu

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$x(t) \xleftrightarrow{\mathcal{L}} X(s)$$

$$X(s) = \mathcal{L}\{x(t)\}$$

$\Rightarrow s = \sigma + jw$ $\longrightarrow$ σ: Reel kısmı

$\hookrightarrow$ w: Sanal kısmı

$$X(\sigma+jw) = \int_{-\infty}^{\infty} x(t)\, e^{-(\sigma+jw)t}\, dt$$

$$= \int_{-\infty}^{\infty} x(t)\, e^{-\sigma t}\, e^{-jwt}\, dt = FF\{x(t)e^{-\sigma t}\}$$

$\Rightarrow s = jw$

$$X(jw) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt = FF\{x(t)\} = X(s)\Big|_{s=jw}$$

s herhangi bir complex sayı olabilir.

(Laplace transformu, $x(t)e^{-\sigma t}$ şeklinde $x(t)$ işaretinin bir reel üstelile çarpımının F.Transformudur

$e^{-\sigma t} \rightarrow \sigma > 0$ zamanda azalan fonk.

$\sigma < 0$ // artan fonk.

Ex: $x(t) = e^{-at}u(t)$

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt = \int_{-\infty}^{\infty} e^{-at}\, u(t)\, e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-(a+s)t}\, dt = \frac{e^{-(a+s)t}}{-(a+s)}\Bigg|_0^{\infty}$$

$$= \frac{1}{s+a}, \quad Re\{s+a\} > 0$$

$s = \sigma + jw \rightarrow X(\sigma+jw) = \dfrac{1}{(\sigma+a)+jw}, \quad \sigma + a > 0$

$s = jw, \sigma = 0 \rightarrow X(jw) = \dfrac{1}{a+jw}, \quad a > 0$

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}, \quad Re\{s\} > -a$$

$a = 0$

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}, \quad Re\{s\} > 0$$

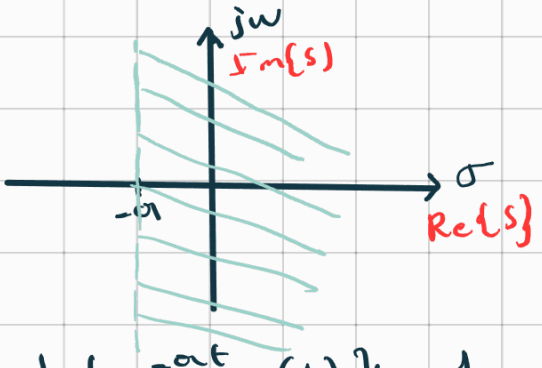
Ex: $x(t) = -e^{-at}u(-t)$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt = -\int_{-\infty}^{\infty} e^{-at}u(-t)e^{-st}dt$$

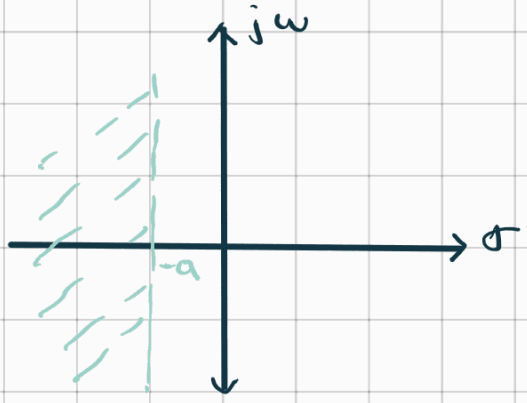
$$= \int_{-\infty}^{0} e^{-(a+s)t}dt = \frac{1}{s+a}, \quad Re\{s+a\} < 0$$

$$-e^{-at}u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \quad Re\{s\} < -a \quad (*)$$

⇒ Yakınsaklık bölgesi (Region of converge, ROC): Laplace transform integralinin yakınsaklığı değer aralığına yakınsaklık bölgesi denir.



$$\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}, \quad Re\{s\} > -a$$



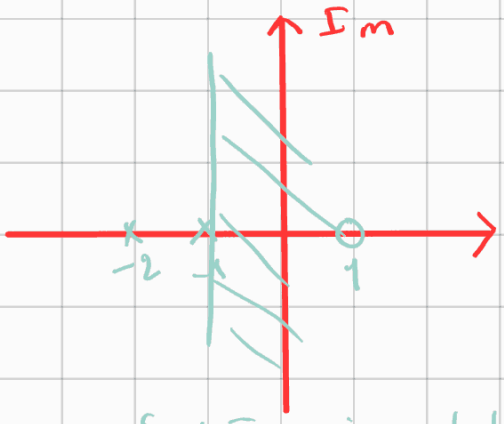
$$\mathcal{L}\{-e^{-at}u(-t)\} = \frac{1}{s+a}, \quad Re\{s\} < -a$$

Ex: $x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$

$$e^{-2t}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+2}, \quad Re\{s\} > -2$$

$$e^{-t}u(t) \longleftrightarrow \frac{1}{s+1}, \quad Re\{s\} > -1$$

$$X(s) = \frac{3}{s+2} - \frac{2}{s+1} = \frac{s-1}{s^2+3s+2}, \quad Re\{s\} > -1$$



kutup
x : pole(

0 : zero
sıfır

s düzlemi kutup sıfır çizimi

Ex: $\delta(t) - \frac{4}{3} e^{-t} u(t) + \frac{1}{2} e^{2t} u(t)$

$$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)\, e^{-st}\, dt = \int_{-\infty}^{\infty} \delta(t) e^{-s0}\, dt$$

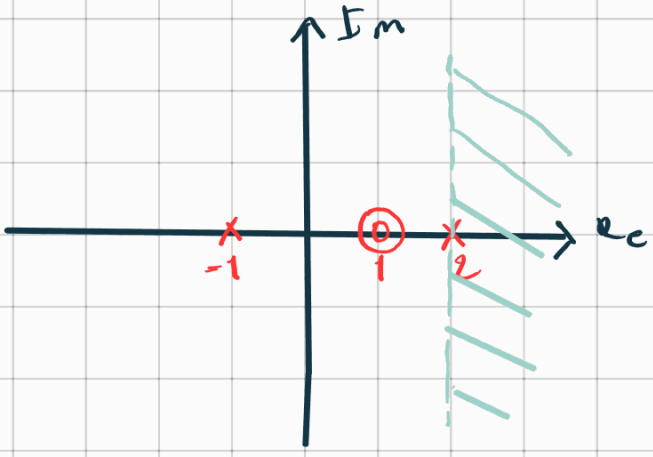
$$= \int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

$\delta(t) \overset{\mathcal{L}}{\longleftrightarrow} 1$, $\forall s$ (Roc, bütün s-düzlemi)

$e^{-t} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+1}$, $Re\{s\} > -1$

$e^{2t} u(t) \longleftrightarrow \frac{1}{s-2}$, $Re\{s\} > 2$

$$X(s) = 1 - \frac{4}{3}\frac{1}{s+1} + \frac{1}{2}\frac{1}{s-2} = \frac{(s-1)^2}{(s+1)(s-2)}$$, $Re\{s\} > 2$



⇒ Aynı noktada tekrar eden s sayısına mertebe denir

Paydada, $s=-1$ ve $s=2$ olmak üzere birinci mertebeden iki adet kutup ve payda $s=1$'de ikinci mertebeden bir adet sıfır vardır.

Not: Laplace transformun ROC'u jw eksenini kesmiyorsa Fourier transformu yakınsamaz.

$\frac{1}{3} e^{2t} u(t)$ ifadesinin Fourier transformu yoktur.

Yakınsama bölgesi (ROC) özellikleri:

1) $X(s)$ in ROC'u s-düzlemindeki jw eksenine paralel şeritlerden oluşur.

$$X(s) = X(\sigma + j\omega)$$

2) Rasyonel Laplace transform için, ROC herhangi bir kutup içermez.

3) $x(t)$ sonlu süreli ve mutlak entegre edilebilir ise, ROC bütün sdüzlemidir

$\int_{T_1}^{T_2} |x(t)|\,dt < \infty$ $\int_{T_1}^{T_2} |x(t)|\, e^{-\sigma t}\, dt < \infty$ ← 1)'deki

→ $Re\{s\} = \sigma = 0$ iken $s = jw$ olup ROC içinde kalır.

→ $Re\{s\} = \sigma > 0$ iken $\int_{T_1}^{T_2} |x(t)|\, e^{-\sigma t} dt < e^{-\sigma T_1} \int_{T_1}^{T_2} |x(t)|\,dt < \infty$ olur ve $Re\{s\} > 0$ s düzlemi ROC içinde kalır.

→ $Re\{s\} = \sigma < 0$ iken $\int_{T_1}^{T_2} |x(t)|\, e^{-\sigma t} dt < e^{-\sigma T_2} \int_{T_1}^{T_2} |x(t)|\,dt < \infty$ dur ve $Re\{s\} < 0$ sdüzlemi ROC içinde kalır.

Ex: $x(t) = \begin{cases} e^{-at}, & 0 < t < T \\ 0, & \text{dışında} \end{cases}$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st}\, dt = \int_0^T e^{-(a+s)t}\, dt = \frac{1}{s+a}\left(1 - e^{-(a+s)T}\right)$$

← 3)'e göre ROC bütün s düzlemi

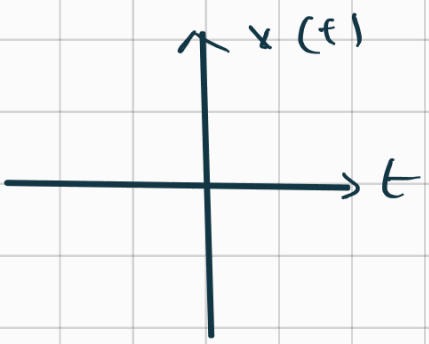
← 2)'e göre ROC kutup içermez, $s = -a$ kutup var.

$s = -a$ iken $X(s) = \frac{0}{0}$

$$\lim_{s \to -a} X(s) = \lim_{s \to -a} \left(\frac{\frac{d}{ds}\left(1 - e^{-(a+s)T}\right)}{\frac{d}{ds}(s+a)} \right)$$

$$= \lim_{s \to a} T e^{-aT} e^{-sT} \Rightarrow x(-a) = T$$

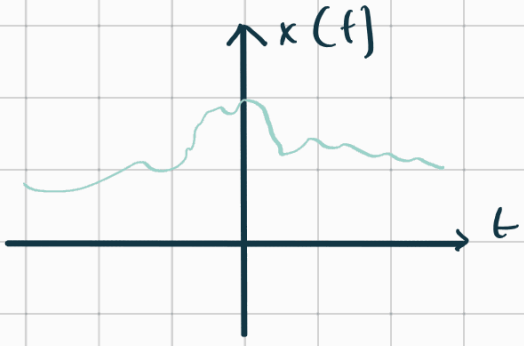
4) $x(t)$ sağ yanlı ve $Re\{s\} = \sigma_0$ doğrusu ROC'daki ise, $Re\{s\} > \sigma_0$ için bütün s değerleri de ROC'da var

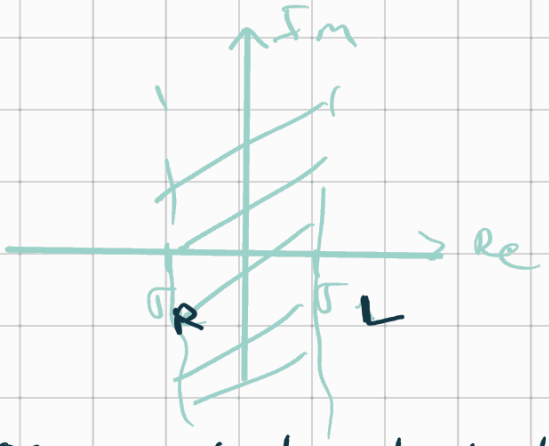




5)

6) x(t) çift yanlı ve $Re\{s\} = \sigma_0$ doğrusu ROC'da ise, ROC, s düzleminde $Re\{s\} = \sigma_0$ doğrusunu içeren bir şeritten oluşur.





7) X(s) rasyonel ise, ROC, ya kutuplarla sınırlıdır yada sonsuza genişler ve kutup içermez.

→ x(t) sağ yanlı ise ROC, en sağdaki kutuların sağında kalan sdüzlemidir

→ x(t) sol yanlı ise, ROC en soldaki kutuların solunda kalan sdüzlemidir

Ex: $x(t) = e^{-b|t|}$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st}\,dt = \int_{-\infty}^{0} e^{bt} e^{-st}\,dt + \int_{0}^{\infty} e^{-bt} e^{-st}\,dt$$

$$= \frac{-1}{s-b} + \frac{1}{s+b}, \quad \begin{matrix} \mathrm{Re}\{s-b\} < 0 \\ \mathrm{Re}\{s+b\} > 0 \end{matrix}$$

$$X(s) = \frac{-2b}{s^2-b^2}, \quad -b < \mathrm{Re}\{s\} < b$$

$x(t) = e^{-b|t|}, b>0$

